

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

Shanmuganathan S R S / 192511347

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

ANNEXURE A – ARTICLE REVIEW

Name: Shanmuganathan S R S **Reg. No.:** 192511347

Selected Research Article

M. Abrishami, S. Dadkhah, E. C. Pinto Neto, P. Xiong, S. Iqbal, S. Ray, and A. A. Ghorbani, "Label Noise Detection in IoT Security based on Decision Tree and Active Learning," in *2022 IEEE 19th International Conference on Smart Communities: Improving Quality of Life Using ICT and AI (HONET)*, Marietta, GA, USA, 2022, pp. 46–53, doi: 10.1109/HONET56683.2022.10019152.

Domain/Application: IoT network intrusion detection and data-centric AI. **ML problem:** detecting and reducing incorrect labels (label noise) in a binary intrusion-detection training dataset using Decision Trees and Active Learning.

Task 1 – Article Summary

The article addresses the problem of label noise in machine-learning datasets used for IoT network intrusion detection. Incorrect labels can reduce the reliability of supervised classification because a model learns from the training labels supplied to it. The authors identify a research gap: although label-noise handling has been studied in machine learning, less attention had been given to label noise in IoT intrusion-detection datasets. They propose a framework combining a robust Classification and Regression Tree (CART) with active learning and uncertainty sampling. First, the decision tree is adjusted so that its minimum number of samples in leaves is selected dynamically, reducing overfitting under noisy labels. Samples with minority labels inside tree leaves are then treated as possible noisy samples. A random forest is used to estimate uncertainty, and uncertain samples are sent for expert review. The reviewed examples are added back to the training set and the process is repeated. The reported experiments show that the active-learning method achieved up to 98% noise reduction for noise levels from 5% to 35%, and 97% and 93% reduction at 40% and 45%, respectively. The study therefore demonstrates that improving training-data quality can be an effective complement to simply increasing model complexity.

Task 2 – Methodology Analysis

Decision Tree/CART: The framework uses a Classification and Regression Tree. The tree separates samples into leaves using feature-based splits and uses majority voting at leaves. Because minority-labelled samples inside a leaf are often inconsistent with the dominant pattern, they can be candidates for label-noise detection. The method dynamically selects the minimum number of samples allowed in a leaf to reduce overfitting.

Active Learning: Uncertainty sampling is used so that a human security expert reviews the most uncertain suspicious samples instead of reviewing the entire dataset. The reviewed samples are used to retrain the model, improving later rounds.

Random Forest: A forest of decision trees is used to obtain attack-versus-benign votes. Samples with small vote differences are considered uncertain and selected for review. The experiments used 500 trees and five vote-difference thresholds: 300, 350, 400, 450 and 500.

Datasets: The study used the BoT-IoT dataset and an IoT Network Intrusion dataset. BoT-IoT originally contains more than 72 million records and 46 features. For experiments, the authors selected a balanced 20,000-sample subset containing 6,481 DoS, 6,482 DDoS, 6,481 Reconnaissance, 79 Theft and 477 benign samples. Six identifying fields were omitted and MinMaxScaler was used to normalize features to 0–1. The second dataset contains Normal, Scanning, MITM, Mirai and DoS traffic. After preprocessing and selection, the experimental dataset contained 100,000 instances. The top 20 features were selected using Mutual Information, and duplicate rows were removed.

CO4 Topic	Mapping to the Article
Decision Trees	CART is the core classifier; leaf size and majority voting are used to make the tree more robust to label noise.
Inductive Learning	The model learns decision rules from labelled training examples and applies those rules to new samples.
Statistical Learning	Classification performance is evaluated using accuracy, precision, recall and F1-score.

Methodological strength: The framework combines an interpretable Decision Tree with uncertainty-based human review, making the approach understandable and reducing unnecessary manual labelling. **Limitation:** the method focuses on symmetric/random label noise and still depends on expert review; therefore, its scalability and generalization to asymmetric noise or multiclass settings require further validation.

Task 3 – Results & Critical Evaluation

The authors compare three noise-detection approaches: Active Learning (AL), Isolation Forest (IF), and Random Forest (RF). The main result is that AL consistently performs best in noise reduction, although it sends more samples for expert review.

Noise level	AL	IF	RF
5%	98%	93%	93%
10%	98%	96%	95%
20%	98%	97%	96%
30%	98%	97%	96%
35%	98%	96%	95%
40%	97%	95%	94%
45%	93%	91%	88%

Interpretation: AL achieved 98% noise reduction from 5% through 35% noise, 97% at 40%, and 93% at 45%. The study also reports that AL was better than IF and RF at preventing wrong relabelling of clean minority samples. Its trade-off was a larger number of samples sent for review.

Critical Evaluation

The reported results are strong because the authors test several injected noise levels and compare multiple methods using the same general experimental setting. The gradual decrease from 98% reduction at lower noise levels to 93% at 45% is also more credible than claiming constant perfect performance. However, the evaluation has limitations. The main experiments use randomly introduced/symmetric label noise, which may not represent all real-world labelling errors. The use of expert review also introduces a human-dependent step and review cost. Furthermore, the approach is mainly demonstrated for binary attack-versus-benign label-noise detection, so its behaviour on more complex multiclass noise should be examined. Stronger evaluation would include naturally noisy datasets with independently verified labels, asymmetric/class-dependent noise, multiple IoT datasets, repeated statistical tests, and comparison with newer robust-loss or modern data-centric learning methods.

Real-World Applicability

The method can support Security Operations Centers, smart-home gateways, IoT intrusion detection systems and network-monitoring platforms. In deployment, it could prioritize suspicious training records for security analysts rather than requiring every record to be manually checked. The main industry challenges are the availability of trustworthy labels, the cost and delay of expert review, changing attack behaviour, class imbalance, privacy of network data, and the need to validate the model on large and continuously changing traffic streams.

Task 4 – Reflection & Learning Outcome

Insight 1 – Decision Tree robustness: I learned that a Decision Tree is not only a classifier; its leaf structure and majority voting can also help identify suspiciously labelled samples. This connects directly to the CO4 Decision Tree topic.

Insight 2 – Active Learning: I learned that uncertainty sampling can reduce manual labelling effort by asking an expert to review only informative samples. This connects classification with practical decision-making.

Insight 3 – Data quality matters: I learned that improving training labels can be as important as changing the ML algorithm. Noise can reduce accuracy and F1-score, so preprocessing and dataset quality are important parts of a machine-learning pipeline.

Proposed Novel Extension

I would extend the research by testing the framework on a modern, multi-class IoT traffic dataset containing several attack families and by replacing the fixed random-noise assumption with class-dependent and time-dependent noise. A second improvement would be to use an adaptive uncertainty threshold so that the number of expert queries changes according to traffic risk and available analyst capacity. This is feasible because the original framework already supports iterative retraining and adjustable thresholds. The expected impact is a more scalable and realistic label-cleaning system for continuously changing IoT environments.

Conclusion

The reviewed IEEE paper demonstrates that Decision Trees and Active Learning can be combined to detect and reduce label noise in IoT intrusion-detection data. Its strongest contribution is the use of interpretable tree structure with

uncertainty-based expert review. The reported noise-reduction results show clear benefits, while the dependence on symmetric noise assumptions and human review indicates important directions for future research.

References / Article Verification

[1] M. Abrishami et al., "Label Noise Detection in IoT Security based on Decision Tree and Active Learning," 2022 IEEE 19th International Conference on Smart Communities: Improving Quality of Life Using ICT and AI (HONET), 2022, pp. 46–53. DOI: 10.1109/HONET56683.2022.10019152.

[2] M. Abrishami, *An Effective Approach to Detect Label Noise*, University of New Brunswick, 2022. The thesis provides the detailed dataset description, preprocessing procedure, algorithms and experimental tables underlying the research.

IEEE Xplore: <https://ieeexplore.ieee.org/document/10019152>

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